Colección de Problemas de Programación Lineal

Problema 1: Compañía de Pinturas

La compañía de pinturas **Manchita** produce tres tipos de pintura adicionando a una pintura base cuatro productos químicos (Q_1 a Q_4).

Se desea determinar la cantidad de toneladas de cada tipo de pintura que debe producirse para maximizar la ganancia total.

Aditivo	Kg de aditivo por tonelada			Disponible (kg)
	Interior	Exterior	Especial	
Q_1	1	2	2	2
Q_2	2	1	1	1
Q_3	1	5	1	3
Q_4	0	0	1	0.8
Ganancia	15000	25000	19000	

Problema 2: Producción en Máquinas

Una empresa produce tres productos procesados en tres máquinas.

Máquina	Tiempo por unidad (min)			Capacidad (min/día)
	Prod.1	Prod.2	Prod.3	
M_1	2	3	2	440
M_2	4	-	3	470
M_3	2	5	-	430

Problema 3: Mezcla de Café

Una casa empacadora recibe diariamente 700 kg de café tipo C y 800 kg tipo K.

Mezcla A:

$$2 \text{ partes C} + 1 \text{ parte K}$$

Ganancia: 22 céntimos/kg.

Mezcla B:

$$1 \text{ parte C} + 2 \text{ partes K}$$

Ganancia: 26 céntimos/kg.

Determinar las cantidades que maximizan la ganancia.

Problema 4: Trenes y Camiones

Un artesano fabrica trenes y camiones.

	Tornillos	Bloques	Ruedas
Tren	10	15	18
Camión	20	10	6
Disponibles	8000	6000	6300

Beneficios:

$$1.6 \text{ €/tren}, \quad 1.4 \text{ €/camión}$$

Problema 5: Calentadores y Aires

Producción diaria con restricciones de horas:

- Calentador: 2h partes + 1h ensamblaje
- Aire acondicionado: 1h partes + 2h ensamblaje

Disponibilidad:

$$8\text{h partes}, \quad 10\text{h ensamblaje}$$

Ganancias:

$$30 \text{ por calentador}, \quad 50 \text{ por aire}$$

Problema 6: Vino y Vinagre

Restricciones:

$$2V \leq G + 4$$

$$3G + 4V \leq 18$$

Beneficio:

$$8V + 2G$$

Problema 7: Tapicería

Tipo A:

$$10m \text{ tela}, \quad 12h$$

Tipo B:

$$15m \text{ tela}, \quad 16h$$

Disponibilidad:

$$300m \text{ tela}, \quad 336h$$

Ganancias:

$$1500A + 2100B$$

Problema 8: Gadgets y Gewgaws

Modelo:

$$\max z = 5x_1 + 4x_2$$

Sujeto a:

$$x_1 + 2x_2 \leq 120$$
$$x_1 + x_2 \leq 70$$
$$2x_1 + x_2 \leq 100$$
$$x_1, x_2 \geq 0$$

Problema 9: Modelo Simple

$$\max 3x_1 + x_2$$

Sujeto a:

$$x_1 + x_2 \leq 7$$
$$x_1 \geq 0$$
$$x_2 \geq 0$$

Problema 10: Dieta de Cerdos

	Corn	Silage	Alfalfa	Requerimiento
Carbs	0.9	0.2	0.4	2
Protein	3	8	6	18
Vitamins	1	2	4	15
Cost	7	6	5	

Problema 11: Modelo Dieta G1 y G2

	Starch	Proteins	Vitamins	Cost
G1	5	4	2	0.6
G2	7	2	1	0.35

Requerimientos diarios:

$$8 \text{ starch}, \quad 15 \text{ proteins}, \quad 3 \text{ vitamins}$$

Problema 12: Giapetto

Modelo:

$$\max Z = \text{Ingresos} - \text{Costos}$$

Restricciones:

$$2F + C \leq 100$$
$$F + C \leq 80$$
$$C \leq 40$$

Problema 13: Industria de Dos Productos

$$\max \text{Profit} = 3x_1 + 4x_2$$

Sujeto a:

$$2x_1 + 3x_2 \leq 16$$
$$4x_1 + 2x_2 \leq 16$$
$$x_1, x_2 \geq 0$$

1 Department Time Constraints

Time used in making x_1 and x_2 units of two products must not exceed the total daily time available in the two shops.

$$2x_1 + 3x_2 \leq 16 \qquad \text{(Machine shop constraint)}$$
$$4x_1 + 2x_2 \leq 16 \qquad \text{(Assembly shop constraint)}$$

Nonnegativity constraints:

$$x_1 \geq 0, \quad x_2 \geq 0.$$

The problem in mathematical form:

$$\textbf{Maximize} \quad Z = 3x_1 + 4x_2$$

subject to

$$2x_1 + 3x_2 \leq 16$$
$$4x_1 + 2x_2 \leq 16$$
$$x_1, x_2 \geq 0.$$

2 Graphical Solution

Consider

$$\max Z = x + 2y$$

subject to

$$x \geq 0, \quad y \geq 0,$$
$$x \leq 6, \quad y \leq 7,$$
$$x + y \leq 12.$$

3 Definition: Linear Programming Problem

A linear programming problem (LP) is an optimization problem where:

1. We maximize or minimize a linear function of decision variables (objective function).
2. The decision variables satisfy a set of linear constraints.
3. Each variable may have sign restrictions ($x_j \geq 0$ or unrestricted).

4 Feasible Region and Optimal Solution

Definition. The feasible region of an LP is the set of all points that satisfy all constraints and sign restrictions.

Definition. For a maximization problem, an optimal solution is a feasible point with the largest objective value. For a minimization problem, it is the feasible point with the smallest objective value.

5 Canonical Linear Programming Problem

Definition (Canonical Form).

1. $x_j \geq 0$, $j = 1, \ldots, N$
2. $Ax = b$
3. Minimize $c^T x$

Matrix form:

$$\min c^T x$$
$$Ax = b$$
$$x \geq 0$$

6 Standard Maximum Problem

Find $x \in \mathbb{R}^n$ to maximize

$$c^T x = c_1 x_1 + \cdots + c_n x_n$$

subject to

$$Ax \leq b$$
$$x \geq 0$$

where

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}$$

7 Standard Minimum Problem

Find $y \in \mathbb{R}^m$ to minimize

$$y^T b$$

subject to

$$y^T A \geq c^T$$
$$y \geq 0$$

8 Extreme Points and Optimality

Observation 1. The optimal solution occurs at an extreme point of the feasible region.

Observation 2. A point is optimal if and only if the negative gradient of the objective function lies in the cone spanned by the gradients of the active constraints.

9 Resource Allocation Model

Let:

$$\max \sum_{j=1}^{n} c_j x_j$$

subject to

$$\sum_{j=1}^{n} a_{ij} x_j \leq b_i, \quad i = 1, \ldots, m$$

$$x_j \geq 0$$

Matrix form:

$$\max z = c^T x$$

$$Ax \leq b$$

$$x \geq 0$$

10 Example 1: The Diet Problem

There are m different types of food $F_1, \ldots, F_m$ that supply n nutrients $N_1, \ldots, N_n$.

Let:

- b_i = price per unit of food F_i,
- a_{ij} = amount of nutrient N_j in one unit of F_i,
- c_j = minimum daily requirement of nutrient N_j,
- y_i = units of food F_i purchased per day.

Cost per day:

$$b_1 y_1 + b_2 y_2 + \cdots + b_m y_m \tag{1}$$

Nutrient requirement:

$$a_{1j} y_1 + a_{2j} y_2 + \cdots + a_{mj} y_m \geq c_j, \quad j = 1, \ldots, n. \tag{2}$$

Nonnegativity:

$$y_i \geq 0, \quad i = 1, \ldots, m. \tag{3}$$

Thus, minimize (1) subject to (2)–(3).

11 Example 2: The Transportation Problem

There are I production plants and J markets.

Let:

- s_i = supply at plant P_i,
- r_j = demand at market M_j,
- b_{ij} = transportation cost from P_i to M_j,
- y_{ij} = amount shipped from P_i to M_j.

Total transportation cost:

$$\sum_{i=1}^{I} \sum_{j=1}^{J} y_{ij} b_{ij}. \tag{4}$$

Supply constraints:

$$\sum_{j=1}^{J} y_{ij} \leq s_i, \quad i = 1, \ldots, I. \tag{5}$$

Demand constraints:

$$\sum_{i=1}^{I} y_{ij} \geq r_j, \quad j = 1, \ldots, J. \tag{6}$$

Nonnegativity:

$$y_{ij} \geq 0. \tag{7}$$

12 Linear Program in Matrix Form

Standard form:

$$\min \sum_{j=1}^{n} c_j x_j$$

subject to

$$\sum_{j=1}^{n} a_{ij}x_j = b_i, \quad i = 1, \ldots, m,$$

$$x_j \geq 0.$$

Matrix representation:

$$\min c^T x \quad \text{subject to} \quad Ax = b, \quad x \geq 0. \tag{LP-1}$$

Where

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad b = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}, \quad c = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix},$$

$$A = \begin{pmatrix} a_{11} & \ldots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \ldots & a_{mn} \end{pmatrix}.$$

13 Basic Terminology

Definition 1. If x satisfies $Ax = b$, $x \geq 0$, then x is *feasible*.

Definition 2. An LP is feasible if it has a feasible solution.

Definition 3. x^* is optimal if

$$c^T x^* = \min\{c^T x : Ax = b, x \geq 0\}.$$

Definition 4. An LP is unbounded if the objective function can decrease without bound over feasible solutions.

14 Equivalent Forms

1. Maximization as minimization:

$$\max c^T x \quad \Longleftrightarrow \quad \min -c^T x.$$

2. Equality as inequalities:

$$a_i^T x = b_i \quad \Longleftrightarrow \quad \begin{cases} a_i^T x \leq b_i, \\ -a_i^T x \leq -b_i. \end{cases} \tag{LP-2}$$

3. Slack variables:

$$a_i^T x \leq b_i \quad \Longleftrightarrow \quad a_i^T x + s_i = b_i, \quad s_i \geq 0.$$

4. Non-positivity:

If $x_j \leq 0$, replace $x_j = -y_j$, $y_j \geq 0$.

5. Unrestricted variable:

$$x_j = x_j^+ - x_j^-, \quad x_j^+, x_j^- \geq 0.$$

15 Geometry of Linear Programming

Let

$$P = \{x : Ax = b, x \geq 0\}.$$

Definition 5. x is a vertex of P if there does not exist $y \neq 0$ such that $x + y \in P$ and $x - y \in P$.

Theorem 1. If $\min\{c^T x : x \in P\}$ is finite, then there exists a vertex solution x^* with $c^T x^* \leq c^T x$ for all $x \in P$.

16 Corollaries

If $\min\{c^T x : Ax = b, x \geq 0\}$ is finite, then there exists an optimal solution which is a vertex.

If $P = \{x : Ax = b, x \geq 0\} \neq \emptyset$, then P has a vertex.

17 Standard Maximum and Minimum Problems

Standard Maximum

$$\max c^T x$$

subject to

$$Ax \leq b, \quad x \geq 0.$$

Standard Minimum

$$\min y^T b$$

subject to

$$y^T A \geq c^T, \quad y \geq 0.$$

Problema 1: Compañía de Pinturas Manchita

La compañía de pinturas **Manchita** produce tres tipos de pintura adicionando a una pintura base cuatro productos o aditivos químicos (Q_1 a Q_4). Se tiene abundante pintura base disponible y su costo ya fue cubierto.

La compañía desea determinar la cantidad de toneladas de cada tipo de pintura que debe producir de manera que maximice la ganancia total. Las únicas restricciones se deben a la disponibilidad de los aditivos químicos requeridos.

Aditivo	**Kg requeridos por tonelada**			**Disponible (kg)**
	Pintura interior	Pintura exterior	Pintura especial	
Q_1	1	2	2	2
Q_2	2	1	1	1
Q_3	1	5	1	3
Q_4	0	0	1	0.8
Ganancia por tonelada	15000	25000	19000	

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Problema 2: Producción en Tres Máquinas

Una empresa produce tres productos procesados en tres máquinas diferentes. El tiempo requerido para fabricar una unidad de cada producto y la capacidad diaria de las máquinas se muestran en la tabla:

Máquina	Producto 1	Producto 2	Producto 3	**Capacidad (min/día)**
M_1	2	3	2	440
M_2	4	-	3	470
M_3	2	5	-	430

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Problema 3: Mezclas de Café

Una casa empacadora recibe diariamente 700 kg de café tipo C y 800 kg de café tipo K. Produce dos mezclas:

- Tipo A: 2 partes de café tipo C y 1 parte de café tipo K. Ganancia de 22 céntimos por kilo.
- Tipo B: 1 parte de café tipo C y 2 partes de café tipo K. Ganancia de 26 céntimos por kilo.

Determinar la cantidad de cada mezcla que debe prepararse para maximizar la ganancia.

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Problema 4: Trenes y Camiones

Un artesano fabrica trenes y camiones de juguete utilizando tornillos, bloques y ruedas.

Disponibilidad semanal:

$$8000 \text{ tornillos}, \quad 6000 \text{ bloques}, \quad 6300 \text{ ruedas}$$

Beneficios:

$$\text{Tren: } 1.6 \text{ euros/unidad} \qquad \text{Camión: } 1.4 \text{ euros/unidad}$$

	Tornillos	Bloques	Ruedas
Tren	10	15	18
Camión	20	10	6

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Example 1.1: Heaters and Air Conditioners

An appliance company manufactures heaters and air conditioners.

- Heater: 2 hours (parts division), 1 hour (assembly division)
- Air conditioner: 1 hour (parts division), 2 hours (assembly division)

Disponibilidad:

$$\text{Parts division: } 8 \text{ hours/day}$$

$$\text{Assembly division: } 10 \text{ hours/day}$$

Profit:

$$\$30 \text{ per heater}, \quad \$50 \text{ per air conditioner}$$

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Industria Vinícola

Una industria produce vino y vinagre.

Restricciones:

$$2(\text{vino}) \leq (\text{vinagre}) + 4$$

$$3(\text{vinagre}) + 4(\text{vino}) \leq 18$$

Beneficio:

$$8 \text{ por vino}, \quad 2 \text{ por vinagre}$$

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Ejercicio No. 0: Tapicería

Tipos de sillones: A y B.

Requerimientos:

- Tipo A: 10 m de tela, 12 horas
- Tipo B: 15 m de tela, 16 horas

Disponibilidad semanal:

$$300 \text{ m tela}, \quad 336 \text{ horas}$$

Utilidad:

$$\$1500 \text{ (A)}, \quad \$2100 \text{ (B)}$$

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Gadgets and Gewgaws

Maximizar:

$$\max z = 5x_1 + 4x_2$$

sujeto a:

$$x_1 + 2x_2 \leq 120$$
$$x_1 + x_2 \leq 70$$
$$2x_1 + x_2 \leq 100$$
$$x_1, x_2 \geq 0$$

—

Ejemplo 1.1

Maximizar:

$$3x_1 + x_2$$

Sujeto a:

$$x_1 + x_2 \leq 7$$
$$x_1 \geq 0, \quad x_2 \geq 0$$

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Pig Diet Problem

Tabla nutricional:

	Corn	Silage	Alfalfa	Requerimiento diario
Carbs	0.9	0.2	0.4	2
Protein	3	8	6	18
Vitamins	1	2	4	15
Cost	7	6	5	

—

Diet Model (G1 y G2)

	Starch	Proteins	Vitamins	Cost ($/kg)
G1	5	4	2	0.6
G2	7	2	1	0.35

Requerimientos diarios:

$$\text{Starch} = 8, \quad \text{Proteins} = 15, \quad \text{Vitamins} = 3$$

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Giapetto's Woodcarving

Variables:

$$x_1 = \text{soldiers}, \quad x_2 = \text{trains}$$

Maximizar utilidad semanal considerando:

- Recursos de acabado y carpintería
- 100 horas de acabado
- 80 horas de carpintería
- Máximo 40 soldados por semana

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General Linear Programming Problem

Maximizar:

$$\text{Max Profit} = 3x_1 + 4x_2$$

Sujeto a:

$$2x_1 + 3x_2 \leq 16$$
$$4x_1 + 2x_2 \leq 16$$
$$x_1, x_2 \geq 0$$

17.1 What is a Linear Programming (LP) Problem?

[Linear Function] A function $f(x_1, x_2, \ldots, x_n)$ of the variables $x_1, x_2, \ldots, x_n$ is a **linear function** if

$$f(x_1, x_2, \ldots, x_n) = c_1x_1 + c_2x_2 + \cdots + c_nx_n$$

where $c_1, c_2, \ldots, c_n$ are constants.

$f(x_1, x_2) = 2x_1 + x_2$ is linear, while $f(x_1, x_2) = x_1^2x_2$ is not linear.

[Linear Inequalities] For any linear function $f(x_1, x_2, \ldots, x_n)$ and any real number b, the inequalities

$$f(x_1, x_2, \ldots, x_n) \leq b \quad \text{and} \quad f(x_1, x_2, \ldots, x_n) \geq b$$

are called **linear inequalities**.

$2x_1 + 3x_2 \leq 3$ is a linear inequality, while $x_1x_2 + \sqrt{x_2} \geq 3$ is not linear.

[Linear Programming Problem] A **linear programming (LP) problem** is an optimization problem in which:

1. We attempt to maximize (profit) or minimize (cost) a linear function (called the *objective function*) of the decision variables.
2. The decision variables must satisfy a set of constraints, and each constraint must be a linear equation or linear inequality.
3. A **sign restriction** is associated with each variable. For any variable x_i, either $x_i \geq 0$ or x_i is unrestricted in sign.

The general structure is:

$$\max \text{ (or min)} \quad z = f(x_1, x_2, \ldots, x_n)$$

$$\text{s.t.} \quad \text{Constraints}$$

$$\text{Sign restrictions}$$

[Example 1.3] Furnco manufactures desks and chairs.

Let:

$$x_1 = \text{number of desks produced}$$

$$x_2 = \text{number of chairs produced}$$

Each desk contributes \$40 profit and each chair \$25. Each desk uses 4 units of wood and each chair 3 units. At most 20 units of wood are available. At least twice as many chairs as desks must be produced.

The LP formulation is:

$$\max \quad z = 40x_1 + 25x_2$$

$$\text{s.t.}$$

$$4x_1 + 3x_2 \leq 20$$

$$x_2 \geq 2x_1$$

$$x_1, x_2 \geq 0$$

Optimal solution:

$$x_1 = 2, \quad x_2 = 4, \quad z = 180.$$

LP Assumptions

1. Proportionality and Additivity

The objective function and constraints must be linear.

- Contributions are proportional to variable values.
- Contributions are independent of other variables.

2. Divisibility Assumption

Decision variables can take fractional values.

If variables must be integers, the problem becomes an **integer programming problem**.

3. Certainty Assumption

All parameters (objective coefficients, RHS values, constraint coefficients) are known with certainty.

17.2 Modeling LP Problems

[Giapetto's Woodcarving]

Let:

$$x_1 = \text{number of soldiers produced per week}$$

$$x_2 = \text{number of trains produced per week}$$

$$\max \quad z = 3x_1 + 2x_2$$

$$\text{s.t.}$$

$$2x_1 + x_2 \leq 100$$

$$x_1 + x_2 \leq 80$$

$$x_1 \leq 40$$

$$x_1, x_2 \geq 0$$

[Farmer Jones]
Method 1: Acres as variables

$$x_1 = \text{acres of wheat}$$
$$x_2 = \text{acres of corn}$$

$$\max \quad z = 100x_1 + 30x_2$$

$$\text{s.t.}$$
$$x_1 + x_2 \leq 7$$
$$10x_2 \geq 30$$
$$10x_1 + 4x_2 \leq 40$$
$$x_1, x_2 \geq 0$$

Method 2: Bushels as variables

$$x_1 = \text{bushels of wheat}$$
$$x_2 = \text{bushels of corn}$$

$$\max \quad z = 4x_1 + 3x_2$$

$$\text{s.t.}$$
$$\frac{1}{25}x_1 + \frac{1}{10}x_2 \leq 7$$
$$\frac{x_2}{10} \geq 3$$
$$\frac{1}{25}x_1 + \frac{1}{10}x_2 \leq 40$$
$$x_1, x_2 \geq 0$$

[Village Butcher Shop]
Let:

$$x_1 = \text{pounds of ground beef}$$
$$x_2 = \text{pounds of ground pork}$$

$$\min \quad z = 80x_1 + 60x_2$$

$$\text{s.t.}$$
$$0.20x_1 + 0.32x_2 \leq 0.25$$

$$x_1 + x_2 = 1$$

$$x_1, x_2 \geq 0$$

[Psychologist Experiment]
Let:

$$x = \text{number of mice}$$

$$y = \text{number of rats}$$

$$\max \quad z = x + y$$

$$\text{s.t.}$$

$$10x + 20y \leq 800$$

$$20x + 10y \leq 640$$

$$x, y \geq 0$$